European Polymer Journal 204 (2024) 112700

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Review

A review of dithieno[3,2-*b*:2′,3′-*d*]pyrrole-based electrochromic conjugated polymers

Buket Bezgin Carbas [a,b,*], Hüseyin Bekir Yildiz [c]

[a] Department of Energy Systems Engineering, Karamanoglu Mehmetbey University, Karaman, Türkiye
[b] Conductive Polymers and Energy Applications Laboratory, Karamanoglu Mehmetbey University, Karaman, Türkiye
[c] Department of Mechanical Engineering, Bartin University, Bartin, Türkiye

ARTICLE INFO

Keywords:
Electrochromic polymers
Electrochromics
Electrochemical polymerization
Electrochemical copolymerization
Dithieno[3
2-*b*:2′
3′-*d*]pyrroles
DTP
Donor–acceptor
Donor-π-acceptor-π-donor

ABSTRACT

Conjugated organic polymers of dithieno[3,2-*b*:2′,3′-*d*]pyrroles (DTPs) have found applications as active materials for electrochromic applications since the chemical structure of DTP monomer gives an opportunity for tuning optoelectronic and electrochemical properties of their corresponding polymer formations. Thanks to versatility of DTP's chemical structure, these properties can be further enriched by N-substitution or cooperation with different aromatic structures in order to synthesize new electrochromic DTP homopolymers or DTP-based copolymers. Although there are many DTP-based polymers in the literature for various technological applications (organic photovoltaics, organic light-emitting diodes, organic field-effect transistors and electrochromics (ECs)), this review only focuses on electrochromic (EC) conjugated polymers (CPs) possessing DTP chemical unit. In this report, we try to review and highlight EC properties of DTP-based polymers in a logical framework. It is aimed that this study will be a reference and inspiration for scientists working on DTP-based ECs, while also providing guidance for practitioners.

1. Introduction

Conjugated polymers (CPs), which impress with their semiconductor properties, are among the popular plastics group with their important features such as lightness, flexibility, easy processing, corrosion resistance, and low production costs. Moreover, CPs obtained by synthetic modifications at the molecular level have the potential to change their specific important properties [1]. Such polymers are used in the energy field (energy storage, energy trapping, and energy translation), flexible electronic devices, and sensors. [2,3]. Challenges and different opinions on this subject maintain their popularity. CPs with various architectures are substantial to build up original substances with superior electrochemical and optical properties for high efficiency and performance in organic electronics such as photovoltaic cells [4,5], light emitting devices [6,7], supercapacitors [8,9], electrochromic devices [10,11], biosensors [12,13] and etc. Depending on functional unit and main chain backbone of CPs, structures and morphologies of CPs influence their electro-optical properties [14]. Designing of organic materials allows new polymeric materials to alter and control redox, electronic, optical, mechanical and processability properties [15].

The source of electrochromic (EC) property in CP structures is related with conduction of charge through alternating single and double bonds through main chain backbone and the color changes depending on applied potential to the material. This provides unique properties to the EC polymer via delocalizing of charges in the structure during doping and dedoping processes via high level of π–orbitals overlapping. During doping process of CPs, new charge carriers (polarons and bipolarons) are formed between frontier molecular orbital energy levels and these charges increase the tendency of the polymer to be conductive. Polaron and bipolaron are radical cation (or anion) and dication (or dianion) states of the polymers, respectively. The polarons are related localized charge carriers associated with a geometrical distortion of the polymer backbone. The bipolaron formation occurs when the two polarons are close together enough leading to a sufficiently large interaction. A dication (dianion) or a bipolaron has two charges associated with the localized polymer segment. The conductivity levels of CPs can be modified by changing doping level. CPs can be doped either by oxidizing (p-type doping) or reducing (n-type doping) the polymer backbone. p-type doping is a type of doping process and occurs during the ejection of an electron from valence band. The injection of an electron to its conduction band is defined as n-type doping. Some polymers have ambipolar character possessing both doping properties. There are many

* Corresponding author.
E-mail address: bcarbas@kmu.edu.tr (B.B. Carbas).

https://doi.org/10.1016/j.eurpolymj.2023.112700
Received 27 November 2023; Received in revised form 13 December 2023; Accepted 18 December 2023
Available online 21 December 2023
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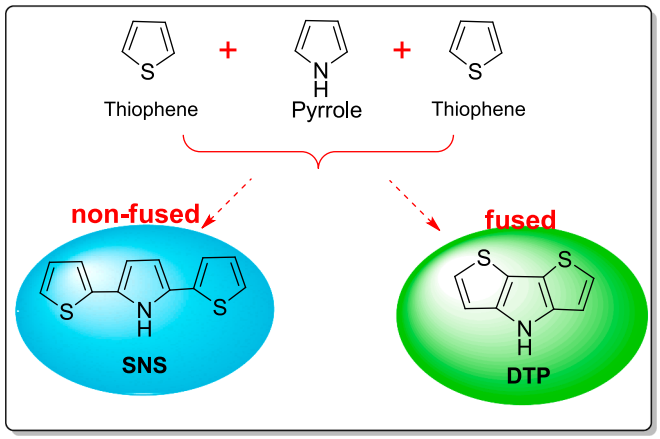
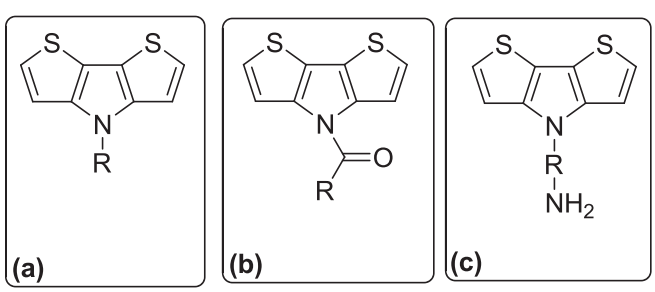


Scheme 1. The chemical structures of thiophene, pyrrole, SNS and DTP.

Table 1
The comparisons of electrochemical and optical properties for P(DTP) and P(SNS) homopolymers.

P	Synt. Meth.	E_m^{ox}(V) [a]	λ_{abs} (nm) [b]	E_{gap} [c] (eV)	EC Properties
P(DTP)	E	0.57 vs. Ag/Ag^+	480	1.80	red (n) blue (ox)
P(SNS)	E	0.87 vs. Ag/Ag^+	430	2.61	yellow orange (n) dark blue (ox)

P: The code of polymer. **Ref**: reference **Synth. Meth.**: the type of synthesis method; **C**: chemical synthesis, **E**: electrochemical polymerization. **ΔT%:** optical contrast %, **CE:** coloration efficiency at λ. E_m^{ox}**(V)**: the first oxidation potential for monomer. λ_{abs}:absorption maximum for polymer E_{gap}: optical bandgap.
[a] the conditions given in in the corresponding references. [b] λ_{abs} in its neutral state. [c] calculated from the onset of the π-π* transition of the neutral state polymer



Scheme 2. Chemical structures of (a) first-generation, (b) second-generation, and (c) new generation DTPs. (R: -alkyl, -aryl, -alkoxy and etc.).

examples of p-doping electrochromic CPs. However, the numbers of n-doped EC polymers are limited in literature [16–18]. The optical band gap (E_{gap}^{opt}) is an electronic transition from highest occupied molecular orbital (HOMO) to lowest unoccupied molecular orbital (LUMO). The band gap and adjustments of energy levels is crucial for CPs. Measuring electrochemical and optical properties is very important in order to characterize a CP. For example, the band gap of CPs can be analyzed with the help of absorption spectroscopy and cyclic voltammetry (CV) / differential pulse voltammetry (DPV), which are optical and electrochemical techniques, respectively. The color determination of CP depends on band energies, the conjugation length and backbone planarity of the CP [19]. The incorporation of electrons affects poor building blocks and integration of fused organic structures in the main chain backbone of CP affect the tuning color properties of material [16–22].

For EC applications, some synthesis strategies (adding functional units to the monomer, achieving copolymerization and Donor (D) -Acceptor (A) approach) are generally used in order to control the optical properties of the resulted material. The reduced band gap and tuned HOMO and LUMO energy levels result in lower oxidation potential, higher absorbance onset, and satisfactory optical contrast for EC polymers. For that reason, great efforts have been made by scientists to get polymers with reduced or desired band gap values. Donor-acceptor (D-A) approach is one of the mostly used strategies. In this approach, a strong donor and an acceptor unit are desired to have a high HOMO energy level and a low LUMO level, respectively. The coupling of these two units allows to obtain a CP with a smaller band gap value than those of both pristine D and A units containing polymers and enhance EC properties of corresponding polymer. Furthermore, the insertion of π-bridges (D- π - A- π - D) in the polymer backbone provides that resulting polymer displays multielectrochromism and ambipolar character with low band gap. Anodically coloring, cathodically coloring and multichromic coloring are types of EC polymers in literature. During polymer switching, EC polymers can have only a color in its oxidation state (anodically coloring EC polymer) and some have only a color in its neutral state (cathodically coloring EC polymer). Multichromic EC polymer, as its name suggests, has different colors at different potentials applied to the polymer film or its redox states [1].

Some parameters (optical contrast (ΔT %), switching time ($t_{\%}$), coloration efficiency (CE), color analysis parameters, stability, and optical memory) are very important to characterize an EC polymer. ΔT % is related with transmittance % difference between colored and bleached state of the polymer at wavelength (λ). $t_{\%}$ can be calculated during the switching in between coloring (t^c) and bleached (t^b) states. This value is calculated for different percentages ($t_{100\%}$, $t_{95\%}$ and $t_{90\%}$) of full optical contrast. CE is formulated by following Equation (1) at a given λ [1,20,21]. ΔOD is determined from %*T* before and after a full switch and is calculated using Equation (2).

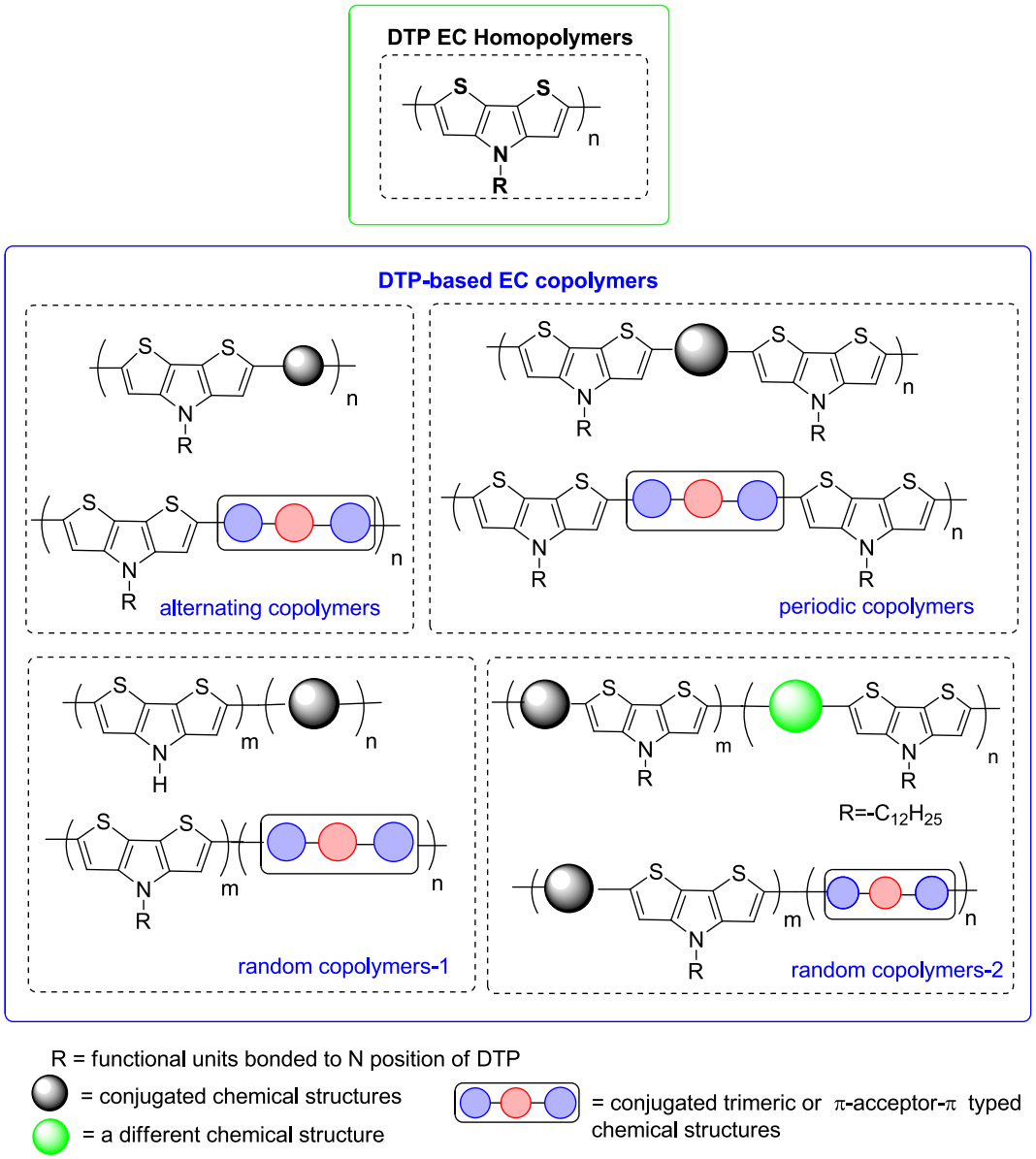
$$CE = \frac{\Delta OD}{Q_d} \tag{1}$$

$$\Delta OD = log\frac{T_b}{T_c} \tag{2}$$

where ΔOD is the difference of optical density, Q_d is the charge density during process, A is the electrode area and T_b and T_c are transmittance values in the bleached and colored states, respectively.

1.1. Polymers of thiophene and pyrrole units Fused/Non-fused formations

The electron-rich conjugated five-membered structures, such as thiophenes, furans pyrroles, and selenophenes, are generally used as donor units in order to get electrochromes with low oxidation potential. Fortunately, these systems are quite amenable to synthetic modification. Among these structures, there are many EC polymers obtained with thiophene and pyrrole possessing different subunits [16,23–25]. The chemical structures of thiophene and pyrrole are shown in Scheme 1. Thiophene and pyrrole-based polymers are shining amongst other CPs. For example, thiophene-based ones are widely used due to their ease of synthesis and stability. Synthetic tailoring on thiophene structure via the incorporation of side chain functionalities offers an opportunity to adjust the electronic and optical properties of the resulting polymer [26–28]. The length of conjugation, which is related to the planarity of the polymer backbone, affects the electro- optical properties of the polymers. Most of the time, side chains are added to the main backbone polymer chain to increase or further increase the processability of the polymer. However, these side chains cause steric interactions (between side chain and side chain or between main backbone and side chain) that can disrupt the planarity of the polymer chain. While forming the polythiophene, the orientational arrangements of the thiophenes with

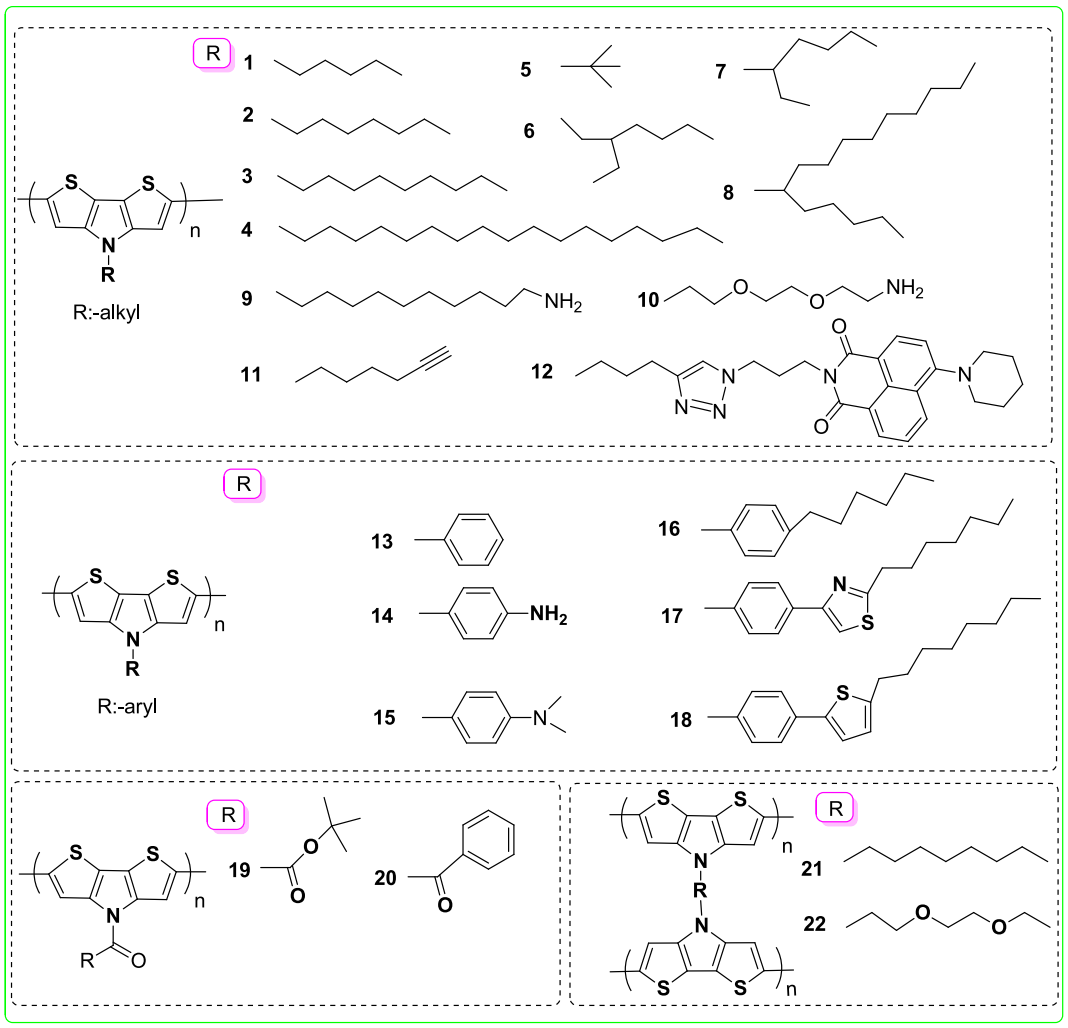


Scheme 3. Classification of DTP-based EC polymers.

added side chains occur in three ways (head-to-head, tail-to-tail and head-to-tail). Polymer chains with head-to-tail combination of side chains have the least steric interaction compared to other sequences. Various methods have been introduced to increase the conjugation of the polymer backbone and thus minimize these undesirable interactions [29]. One of the methods used to make the properties of the polymer more effective is the cyclization of repeating aromatic rings. Fusion of the rings increases the structural rigidity of the main chain by reducing the rotational motion between the rings. Such a formation increases the planarity of the system and ensures a better overlap of π-orbitals for polymer. For example, the interlocking of two thiophene rings reduces the interring distance compared to that of the thiophene unit, and this causes the quinoid resonance of the polymer to be more stable. For that reason, incorporation of fused thiophene derivatives into the backbone of many CPs has become very popular lately. Since this rigid and coplanar structure enhances effective π-conjugation, it lowers the band gap and extends the absorption with intriguing EC properties [30–33].

Fused bithiophene based polymers containing different heteroatoms such as nitrogen, silicon, phosphorous and germanium have been combined with various donor acceptor units and striking performance results have been obtained in electrochromic and optoelectronic devices [34–39]. One of the fused ring systems of thiophene is formed with dithieno[3,2-*b*:2′,3′-*d*] thiophene (DTT). The number of active ends possible for functionalization of this structure is little. An alternative to this structure and with superior functionality, dithieno[3,2-*b*:2′,3′-*d*] pyrrole (DTP) structure, has been put forward by scientists (Scheme 1). While such a chemical structure provided the system with a heteroatomic bridge function, it also paved the way for N-functionalization [40]. Another popular platform that will increase the N-functionality of pyrrole and non-fused structure of DTP is 2,5-di(thienyl)pyrroles (SNS) (Scheme 1). This trimeric non-fused structure is formed by bonding the 2 and 5 positions of pyrrole to the one of each α-positions of two thiophene structures. In literature, there are many examples of different N-substituted SNS EC polymers and their copolymer formations with 3,4-ethylenedioxythiophene (EDOT) [20,21].

The fused ring structure of DTP improves conjugation and increases planarity leading to a bathochromic shift of absorption maximum. This situation gives a hint that the EC activity of DTP-based homopolymers and copolymers will be more effective in the near infrared region of spectrum in their oxidized states. Therefore, its polymers can find



Scheme 4. Chemical structures of DTP EC homopolymers.

applications in the field of smart windows, optical fiber communications systems and photodetectors. The oxidation potential of DTP monomer (0.57 V vs. Ag/Ag^{+}) has a lower value than that of SNS unit (0.87 V vs. Ag/Ag^{+}) since fused ring system enhances more donor effect with its π-extended structure [20,40]. When the monomers are polymerized, homopolymer P(DTP) has lower band gap (1.8 eV) and higher absorption maximum value (480 nm) than those (2.61 eV and 430 nm) of P (SNS) [20,40–42] (Table 1). As a result, P(DTP) gives a color that shifts towards the lower energy region of the spectrum in its neutral state.

Unfunctionalized monomer DTP was firstly synthesized by Zanirato and coworkers [43]. Later, many research groups were synthesized new DTP derivatives with different subunits. Rasmussen and coworkers reviewed and classified DTP monomers into two groups depending on type of their side chains. Rasmussen et al., most recently in 2013 mentioned the synthesis and applications of DTP-based polymers in detail . However, no detailed study on its electrochromic properties has been mentioned [40]. Therefore, in this study, we wanted to emphasize in detail the electrochromic applications of such polymers, which are the missing piece of the puzzle. Although it is seen as a ghost structure, interest in DTP-based polymers with electrochromic properties has been increasing day by day. While the first-generation DTPs possess traditional alkyl or aryl side chains, the second-generation ones have electron withdrawing acyl side chains at their functional active N-functional unit [40] (Scheme 2). Then other research groups added new DTP monomers with new side chains which are called new generation DTP systems [44–47].

1.2. First studies related EC properties of Poly(DTP)

The first applications of DTP chemical structure into CPs were done by Zotti and coworkers [41]. The monomers (unfunctionalized DTP and some N-alkyl functionalized DTPs) were electropolymerized on 0,8 x 2,5 cm indium-tin-oxide (ITO) sheet in the medium of 0.1 M tetraethyl ammonium perchlorate (TEAP) /acetonitrile (ACN) via repetitive cyclic voltammetry (CV) method. The first cyclic voltammogram of unfunctionalized DTP gave an oxidation peak at about 0.52 V vs. Ag/AgCl. The spectroelectrochemical studies were done for Poly (DTP) and gave a strong band at 480 nm in its neutral state. Upon oxidation of polymer film, the intensity of this band decreased, and a new intermediate band appeared at about 700 nm. However, no details related to their EC properties were given. The experimental results also showed that Poly (DTP) had a band gap of ~ 1.7 eV [41]. The first EC properties of Poly (DTP) were analyzed by Rasmussen and coworkers [42]. The polymer was electrochemically synthesized on ITO electrode in the medium of 0.1 M tetrabutylammonium hexafluorophosphate ($TBAPF_6$) / acetonitrile (ACN). Unlike the spectroelectrochemical studies obtained by Zotti et al., polymer showed an extra major peak at about 1650 nm (a higher intensified peak than that of 700 nm) in its oxidized state [41,42]. Rasmussen and coworkers gave information about the colors of neutral and oxidized states of polymer film. Poly (DTP) had a color transition between red (in the neutral state) and blue (in the oxidized state) [42].

Table 2
Some EC properties of DTP homopolymers in literature.

P [ref]	Synt. Meth.	E_m^{ox}(V) [a]	λ_{abs} (nm) [b]	E_{gap} [c] (eV)	EC Properties [d,e] (L/a/b)
1 [49]	E	0.51 vs. Fc/Fc^+	481	1.82	blue (n) colorless (ox)
2 [42]	E	0.56 vs. Ag/Ag^+	545	1.71	reddish brown (n) blue (ox)
3 [42]	E	0.56 vs. Ag/Ag^+	546	1.70	reddish brown (n) blue (ox)
4 [42]	E	0.56 vs. Ag/Ag^+	555	1.68	reddish brown (n) blue (ox)
5 [42]	E	0.57 vs. Ag/Ag^+	510	1.70	reddish brown (n) blue (ox)
6 [42]	E	0.56 vs. Ag/Ag^+	600	1.64	reddish brown (n)blue (ox)
7 [49]	E	0.50 vs. Fc/Fc^+	558	1.82	blue (n)colorless (ox)
8 [49]	E	0.50 vs. Fc/Fc^+	550	1.81	dark blue (n) colorless (ox)
9 [44]	E	1.30 vs. Ag wire	475	2.00	$t_{95\%}$ [c] (s) = 0.7 (475 nm) / 0.9 (1030 nm), ΔT% =35(475 nm) / 50 (1030 nm), dark yellow(0.0 V) (73.74/13.81/6.92), gray (0.5 V) (70.11/-15.21/9.06), blue(1.1 V) (75.89/-24.25/1.28)
10 [46]	E	1.27 vs. Ag wire	420	1.92	$t_{95\%}$ [c] (s) = 1.8 (420 nm) / 1.0 (770 nm) / 0.8 (980 nm), ΔT% =9.0(1.1 V) (75.89/-24.25/1.28), reddish orange(0.0 V) (108.2/71.3/44.5), brown(0.6 V) (68.8/55.5/38.7), dark blue(1.2 V) (64.4/-55.1/43.3)
11 [45]	E	0.80 vs. Ag/AgCl	703	1.76	yellow (-0.5 V) / blue (0.9 V)
12 [45]	E	0.70 vs. Ag/AgCl	680	1.82	red (-0.5 V)/blue (0.9 V)
13 [49]	E	0.60 vs. Fc/Fc^+	439	1.92	dark blue (n) colorless (ox)
14 [44]	E	1.70 vs. Ag wire	440	1.65	$t_{95\%}$ [c] (s) = 1.8 (440 nm) / 1.5 (1060 nm), ΔT% =14 (440 nm) / 32 (1060 nm), brown (0.0 V) (67.87/-1.56/37.94), blue (1.2 V) (72.10/–22.32/-0.02),
15 [50]	E	0.92 vs. Ag/AgCl	478	2.03	$t_{95\%}$ [c] (s) = 1.68 (470 nm)/0.53 (670 nm)/0.62 (1100 nm), $t_{95\%}$ [b] (s) = 0.5 (470 nm)/4.42 (670 nm)/5.53(1100 nm), ΔT% =23 (1.2 V) (72.10/–22.32/-0.02), CE(cm^2/C) = 84(470 nm)/95 (670 nm)/59(1100 nm), orange (n), light blue (ox)
16 [42]	E	0.65 vs. Ag/Ag^+	515	1.73	reddish brown (n)/blue (ox)
17 [51]	E	0.63 vs. Fc/Fc^+	502	1.78	brown (-0.4 V)/ dark brown (0.0 V) / dark blue (0.4 V)/ blue(0.6 V)
18 [51]	E	0.63 vs. Fc/Fc^+	519	1.76	green (-0.37 V) / brown (-0.15 V) / reddish brown (0.0 V) / yellowish green(0.5 V)/)green (0.7 V)
19 [49]	E	0.74 vs. Fc/Fc^+	388	1.96	dark blue (n) / colorless (ox)
20 [49]	E	0.73 vs. Fc/Fc^+	481	2.07	dark blue (n)/colorless (ox)
21 [44]	E	1.10 vs. Ag wire	490	1.8	$t_{95\%}$ [c] (s) = 0.8 (490 nm) / 0.5 (680 nm) / 0.8 (1040 nm), ΔT% =45(490 nm) / 24 (680 nm) / 56 (1040 nm), red(0.0 V) (46.58/53.8/36.94), dark brown(0.2 V) (50.22/23.55/34.86), light gray(0.3 V) (53.49/-11.36/13.63), blue(0.5 V) (63.01/-29.65/-6.53)
22 [47]	E	1.00 vs. Ag wire	450	1.82	$t_{95\%}$ [c] (s) = 1.2 (450 nm) / 0.7 (710 nm) / 0.8 (900 nm), ΔT% =30(490 nm) / 24 (680 nm) / 56 (1040 nm), red(0.0 V) (59.8/11.6/17.7), brown(0.5 V) (63.5/11.3/19.7), gray(0.7 V) (66.9/-3.6/8.7), blue(1.1 V) (65.8/-4.4/-2.33)

P: The code of polymer. **Ref**: reference **Synth. Meth.**: the type of synthesis method; **C**: chemical synthesis, **E**: electrochemical polymerization. **E_m^{ox}(V)**: the first oxidation potential for monomer. **λ_{abs}**:absorption maximum for polymer. *predicted data from reference given, red1: first reduced state, red2: second reduced state, ox1: first oxidized state, ox2: second oxidized state.

[a] the conditions given in in the corresponding references).

[b] λ_{abs} in its neutral state.

[c] calculated from the onset of the π-π* transition of the neutral state polymer.

[d] the coloring switching time (at λ) at $t^c_{100\%}$: 100%, $t^c_{95\%}$: 95%, $t^c_{90\%}$: 95% of ultimate contrast, the bleaching switching time at $t^b_{100\%}$: 100%, $t^b_{95\%}$: 95%, $t^b_{90\%}$: 90% of ultimate contrast. If percentages or colored/bleached states (at a given λ) are not written, time is only given as t%.

[e] colors at a given applied potential or states; n: neutral state, i: intermediate state, ox: fully oxidized state.

1.3. Classification of DTP-based EC polymers

DTP-based EC polymers can be divided into two classes: DTP EC homopolymers and DTP-based EC copolymers. The copolymers can be examined into three groups; DTP-based periodic(-ABA-ABA-), alternating (-ABAB-), and random copolymers depending on the arrangement of DTP unit between new unit through the main chain backbone (Scheme 3). For CPs showing EC property, optical and electrochemical properties of the corresponding polymer are very important besides band gap control. These properties can be modified with molecular design or arrangement of polymer chain. The adjustments of frontier orbital energies can be applied in three ways [48]. One of them is to enlarge the π orbital system. The other one is connected with the first one and uses planar fused aromatic ring systems in the main chain backbone. The last one is related to the arrangement of donor (D) and acceptor (A) units along the main chain. HOMO level for polymer structure increase can be observed for the first two. D-A typed EC CPs have been very popular for various organic electronic applications lately. For that reason, the control of energy levels and thus the bandgap of CP is very crucial. The determination of HOMO and LUMO level can

Table 3
Some DTP-based EC alternating copolymers in literature.

P *[ref]*	Synt. Meth.	λ_{abs} (nm) [b]	E_{gap} [c] (eV)	EC Properties [d,e] (L/a/b)
23 [52]	C	470	1.41	blue (ox)/green (n)/light purple (red1)
24 [52]	C	510	0.87	gray (ox) /dark pink (n) /darker purple (red 1)/bright blue (red 2)
25 [52]	C	517	0.56	blue (ox) /purple (n) /darker purple (red 1) /bright purple (red 2)
26 [53]	C	500	1.72	purple (n), black (i), sky blue(ox), $t_{?\%}$ (s) = 0.8 (542 nm) / 2.5 (765 nm) / 2.4 (1345 nm), ΔT% =42(542 nm) / 26.5 (765 nm) / 70 (1345 nm), CE(cm^2/C) = 153 (542 nm) / 240 (765 nm) / 85 (1345 nm)
27 [53]	C	490	1.70	red (n), purple (i1), brown (i2), black (i3), green (i4), sky blue (ox), $t_{?\%}$ (s) = 0.5 (545 nm) / 1.9 (785 nm) / 0.5 (1245 nm), ΔT% =27(545 nm)/36(785 nm)/49(1245 nm), CE(cm^2/C) = 611(545 nm)/447(785 nm)/920(1245 nm)
28 [53]	C	508	1.69	red (n), orange brown (i1), purple (i2), gray (i3), greenish blue (i4), blue (ox), $t_{?\%}$ (s) = 1.8 (530 nm) / 2.9 (765 nm) / 2.4 (1345 nm), ΔT% =32(530 nm) / 45 (1330 nm), CE(cm^2/C) = 120 (530 nm) / 157 (1330 nm)
29 [54]	C	630	1.31	$t_{?\%}$ (s) = 1.2 (400 nm) /2.3 (1100 nm), ΔT% =26.7(400 nm) /53.4 (1100 nm), CE (cm^2/C) = 36.83 (400 nm) / 91.33 (1100 nm)
30 [54]	C	645	1.29	$t_{?\%}$ (s) = 2.8 (400 nm) / 3.8 (1100 nm), ΔT% =8.1 (400 nm) / 3.4 (1100 nm), CE(cm^2/C) = 7.68 (400 nm) / 10.92 (1100 nm)

P: The code of polymer. **Ref**: reference **Synth. Meth.**: the type of synthesis method; **C**: chemical synthesis, **E**: electrochemical polymerization. λ_{abs}: absorption maximum for polymer.
*predicted data from reference given, red1: first reduced state, red2: second reduced state, ox1: first oxidized state, ox2: second oxidized state.
[b] λ_{abs} in its neutral state.
[c] calculated from the onset of the π-π* transition of the neutral state polymer.
[d] the coloring switching time (at λ) at $t^c_{100\%}$: 100%, $t^c_{95\%}$: 95%, $t^c_{90\%}$: 95% of ultimate contrast, the bleaching switching time at $t^b_{100\%}$: 100%, $t^b_{95\%}$: 95%, $t^b_{90\%}$: 90% of ultimate contrast. If percentages or colored/bleached states (at a given λ) are not written, time is only given as t%.
[e] colors at a given applied potential or states; n: neutral state, i: intermediate state, ox: fully oxidized state.

be done with the selection of donor and acceptor unit, respectively. The band gap control also helps to optimize the color change for EC device applications [1].

1.3.1. EC properties of DTP homopolymers

The chemical structures of EC DTP homopolymers are summarized in Scheme 4. The Rasmussen research group functionalized the N position of DTP with alkyl units of various lengths [28]. They polymerized the monomers both chemically and electrochemically. The polymers in question, **1**, **2**, **3**, **4**, **5**, **6** and **16** were unfortunately partially soluble and gave red emission in the solution. They all show the same EC behavior just like unsubstituted P(DTP) polymeric film.

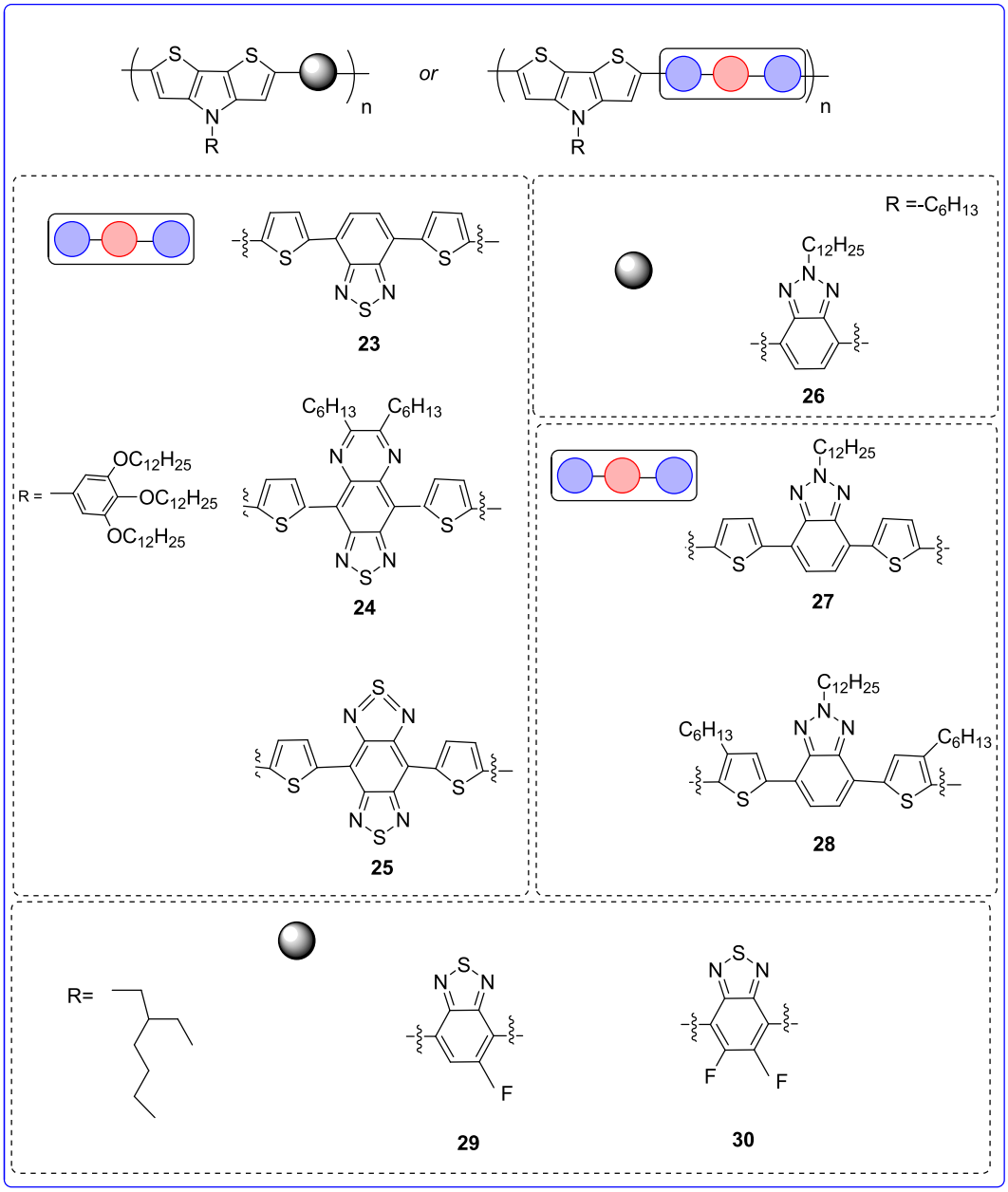
Bäuerle research group synthesized and electropolymerized new N-functionalized analogues with alkyl units (**1**, **7**, **8**, **13**) and –acyl units (**19**, **20**) [49]. All polymers exhibited EC behavior between dark blue to colorless upon oxidation. Alkyl- (**9**) and alkoxy- (**10**) amine functionalized P(DTP) EC films were also electropolymerized and showed approximately same color tones upon oxidation with two different intermediate colors [44,46]. An N-akynylated DTP polymer (**11**) with a terminal triple bond was synthesized. Its monomer was clicked with azide containing organic compound and polymer **12** was achieved. While both polymers show blue colors in their oxidized states, yellow (for polymer **11**) and red (for polymer **12**) colors were observed in their neutral states [45]. N-aryl functionalized DTP polymers (**13**, **14**, **15**, **16**, **17**, **18**) were also synthesized and their EC properties were investigated [42,44,49–51]. N-acyl substituted DTP polymers (**19**, **20**) show EC property with the colors of dark blue to colorless upon doping [49]. New generation EC polymers (**21**, **22**) were prepared with alkyl – and alkoxy-bridged DTP monomers and the results showed that both polymers had more color variation during oxidation than other Poly(DTP) derivatives [44,47].

It was found that the E^{ox}_m of alkyl-functionalized DTP monomers (monomer of **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**) is not higher than 0.6 V vs. Ag/Ag^+ like unfunctionalized DTP monomer, whereas the oxidation potential of the amine or bulky -terminated alkyl, acyl and bridged functionalized monomers is more positive oxidation potentials. Furthermore, the optical band gaps of homopolymers are between 1.6 eV and 2.0 eV as seen Table 2. There is not a relationship between electron donor or acceptor effect of N-functionalized unit on monomer oxidation potential and band gap values of homopolymers. Among the polymers in Table 2, EC properties of homopolymers **9**, **14**, **15**, **17**, **18**, **21** and **22** were studied in detail.

1.3.2. DTP-based EC copolymers

The experimental results in literature showed that DTP building block enhanced charge carrier mobility, solubility, solid state fluorescence with reduced and low band gaps when it was incorporated into the polymer chain [40]. Although DTPs were used as electron donor units, they had the disadvantage of weak electron transport through the main chain backbone. Some conjugated units possessing electron acceptor property or acceptor units linked with two π donor units were added into DTP units in order to get low band gap copolymer. In this part of the review, DTP-based copolymers possessing EC behavior were discussed.

1.3.2.1. DTP-based EC alternating copolymers. During the synthesis of DTP-based EC alternating copolymers, usually hexyl, 3,4,5-tri-ndecyloxyphenyl and 3-heptyl ethyl units were studied. All the polymers were chemically synthesized as seen in Table 3. N-(3,4,5-tri-ndecyloxyphenyl) functionalized DTP monomer was incorporated a ternary structure containing two π-linker (thienyl) and one acceptor unit in order to obtain copolymers **23**, **24** and **25**. Three different acceptor units; 4,7-dithien-2-yl-[2,1,3]-benzothiadiazole, 4,9 dithien-2-yl-6,7-di-*n*-hexyl-[1,2,5]thiadiazolo[3,4-g]quinoxaline and 4,8-dithien-2-yl-2 λ^4 σ^2-benzo[1,2- *c*;4,5-*c*']bis[1,2,5]thiadiazole were used for this purpose by Marder research group [52] (Scheme 5). All copolymers displayed ambipolar character in their electrochemical behavior with three or



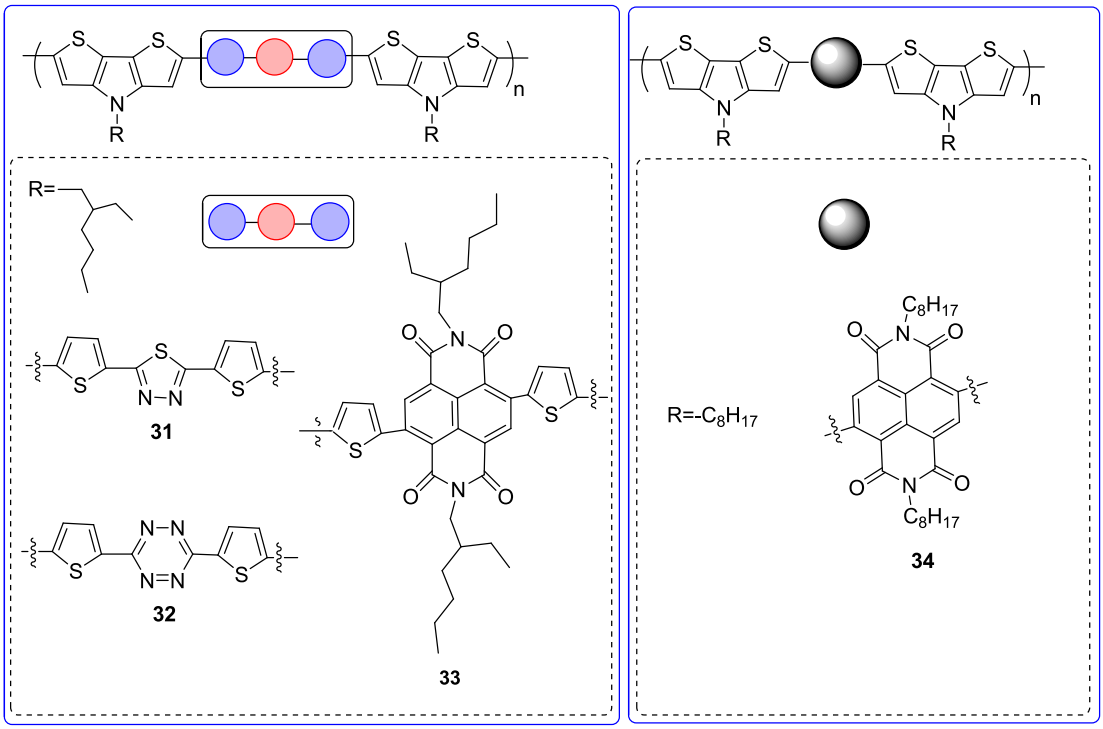
Scheme 5. Chemical structures of DTP-based EC alternating copolymers in literature.

more electrochromic colors as shown in Table 3.

A typical donor group; N-alkyl functionalized DTP unit was incorporated to two different ternary structure including two different π-linkers (thienyl / 3- hexylthienyl) and same acceptor unit (2-dodecyl-4,7-di(thiophen-2-yl)–2H-benzo[d][1,2,3] triazole) [53] in order to compare the influence of different π-linkers on electro-optical properties of resulted copolymers (**27** and **28**) (Scheme 5) [53]. These two alternating copolymers were chemically obtained by Stille coupling. Cirpan research group also tested how these properties besides EC properties of the structures with or without π-bridges would change. For this purpose, copolymer **26** was also synthesized and characterized under the same conditions as the other copolymers in the study [53]. It has been revealed that the energy levels, CE, switching time and optical contrast values of copolymers **27** and **28** (with different π-linker structures) are different but it was observed that the band gap of the copolymers was not affected much. While **27** and **28** showed electrochemical ambipolar character, **26** had only p-type doping properties. Depending on doping type, each polymer showed different colors in their redox states as seen in Table 3. Kinetic studies related showed that **27** and **28** had outstanding CE and ΔT% values in the near infrared region. Among them, copolymer **27** had low switching time since it created less steric effect allowing ion diffusion and dopant insertion in the structure compared to copolymer **28**.

As seen in the next example, the Cirpan group tried to investigate the effect of increasing the F number in the acceptor group on the EC properties of the resulting DTP-based alternating copolymer [54]. N-alkylated functionalized unit as donor unit was incorporated to two different acceptor unit with different numbers of F substitutions in order to get **29** and **30** (Scheme 5). The spectroelectrochemical experiments showed that difluorinated copolymer had a red shifted absorption causing a lower band gap. It was found that **30** possessed a lower ΔT% values than that of **29** (Table 3).

1.3.2.2. DTP-based EC periodic copolymers. The chemical structures of periodic copolymer formations were displayed in Scheme 6 and their EC properties were summarized in Table 4. Three donor-π-acceptor-



Scheme 6. Chemical structures of DTP-based EC periodic copolymers in literature.

Table 4
Some DTP-based EC periodic copolymers in literature.

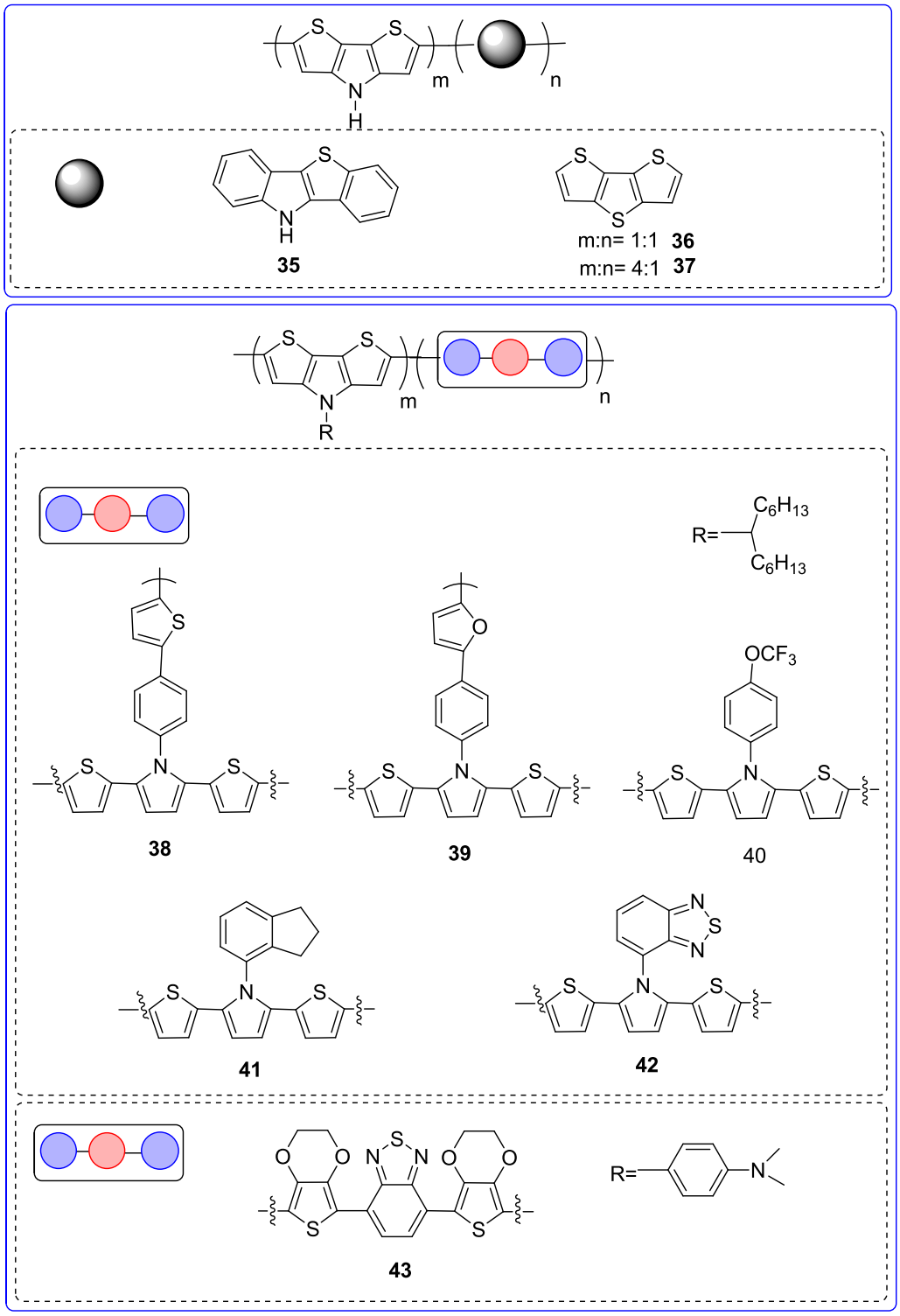
P *[ref]*	Synt. Meth.	λ_{abs} (nm) [b]	E_{gap} [c] (eV)	EC Properties [d,e] (L/a/b)
31 [55]	**E**	–	1.71	(red1) 70/34/21, (n) 61/18/4, (ox1)60/13/-4, (ox2)70/2/-1
32 [55]	**E**	–	1.55	(red2) 41/-8/43, (red1) 43/21/41, (n) 45/27/31, (ox1)46/-20/-17, (ox2)51/0/-17
33 [55]	**E**	–	0.84	(red1) 90/-12/-7, (n) 6/14/-3, (ox1)78/-8/-18, (ox2)85/-14/-19
34 [56]	**E**	468	1.65	green* (-1.65 V), reddish brown* (-1.12 V), greenish yellow* (-0.85 V), green* (0.11 V), dark blue* (1.0 V)

P: The code of polymer. **Ref**: reference **Synth. Meth.**: the type of synthesis method; **C**: chemical synthesis, **E**: electrochemical polymerization. λ_{abs}: absorption maximum for polymer.
[b] λ_{abs} in its neutral state.
[c] calculated from the onset of the π-π* transition of the neutral state polymer.
[d] the coloring switching time (at λ) at $t^{c}_{100\%}$: 100%, $t^{c}_{95\%}$: 95%, $t^{c}_{90\%}$: 95% of ultimate contrast, the bleaching switching time at $t^{b}_{100\%}$: 100%, $t^{b}_{95\%}$: 95%, $t^{b}_{90\%}$: 90% of ultimate contrast. If percentages or colored/bleached states (at a given λ) are not written, time is only given as t%.
[e] colors at a given applied potential or states; n: neutral state, i: intermediate state, ox: fully oxidized state.
[*] predicted data from reference given, red1: first reduced state, red2: second reduced state, ox1: first oxidized state, ox2: second oxidized state.

π-donor based copolymer formation were obtained with three different acceptor units (1,3,4-thiadiazole, 1,2,4,5-tetrazine (TZ) and benzo[lmn][3,8]-phenanthroline-1,3,6,8(2H,7H)-tetraone. N-alkyl functionalized DTP and 2,5-thienylene were used as same donor and π-linker structures in the linkage, respectively (Scheme 6). **31**, **32** and **33** copolymers were electrochemically obtained with cyclic voltammetry method and their EC colors were reported. All the copolymers were electrochemically synthesized. They provided a rich color palette with four or five redox states not only in the visible but also in the near infrared region [55]. The copolymers **32** and **33** showed strong ambipolar character leading to lower band gaps than that of copolymer **31** as shown in Table 4. Higher optical contrast differences were observed in the near part of infrared region than in the visible region.

The naphthalene bisimide unit was symmetrically core functionalized with N-alkyl functionalized DTP unit in order to obtain D-A-D type copolymer (**34**) without the presence of a linker [56] (Scheme 6). Copolymer **34** showed ambipolar character in its electrochemical behavior with four reversible redox couples and a low band gap (1.65 eV). Furthermore, the copolymer **34** had different colors (green (at −1.65 V), reddish brown (at −1.12 V), greenish yellow (at −0.85 V), green (at 0.11 V), dark blue (at 1.0 V) at different potentials (Table 4).

1.3.2.3. DTP-based EC random copolymers. DTP monomer was electrochemically copolymerized with thionaphtheneindole and dithienothiophene with different molar ratios and their corresponding random copolymers; **35** (m:n = 3:1) and **36** (m:n = 1:1) and **37** (m:n = 4:1), were obtained, respectively [57,58] (Scheme 7). While all copolymers mentioned exhibit EC behavior, no details related with copolymer color or other EC parameters were given. After these random copolymers, ternary SNS group functionalized different subunits was electrocopolymerized with alkyl group functionalized DTP monomer by Wu and coworkers [59–62]. DTP-based copolymers of phenyl thiophene (**38**), phenyl furan (**39**), phenyl trifluoromethoxy (**40**), indene (**41**), and benzothiadiazole (**42**), containing SNS structures were obtained ((Scheme 7) and their EC properties were reported. EC parameters changed depending on different side of functional unit in the copolymer formation. The colors of all copolymers were the tones of yellow (in the neutral state) and blue (in the oxidized state). They also have colors in their intermediate states (Table 5). The experimental results showed that ΔT% and CE values of copolymers increased significantly in the near-infrared region. Dual type ECDs were designed with these copolymers as anodically coloring materials and poly(3,4-(2,2-diethylpropylenedioxy)thiophene) (PProDOT-Et_2) as cathodically coloring



Scheme 7. Chemical structures of DTP-based EC random copolymers-1 in literature.

material [59–62]. All ECDs displayed fast switching time (≤2.0 s) and long-term optical stability (higher than 95 %).

N-dimethylaniline functionalized DTP monomer was electrochemically incorporated to 4,7-di(2,3-dihydro-thieno[3,4-b][1,4]dioxin-5-yl) benzo[1,2,5]thiadiazole (BDT) and the random copolymer (**43**) was obtained by Hu (Scheme 7) [50]. The entire visible region was almost covered by this copolymer in its neutral state and thus it displayed a black color. ΔT% and CE values were found as 41 and 236 cm^2/C at 1100 nm, respectively (Table 5).

Two different electron acceptor units (perylene diimide and benzotriazole) possessing random copolymer with a DTP π-bridged structure was synthesized (**44**) by Toppare research group [63] (Scheme 8). They tried to analyze the effect of π-bridge on the electrochemical and spectroelectrochemical properties of **44** [63]. The electrochemical experimental results showed that **44** polymer was soluble in chloroform and had an ambipolar character with different colors (black (during p-doping), brown (in the neutral state) and blue (during n-doping) at different doping levels). DTP units as π-bridge role in the structure

Table 5
Some DTP-based EC random copolymers in literature.

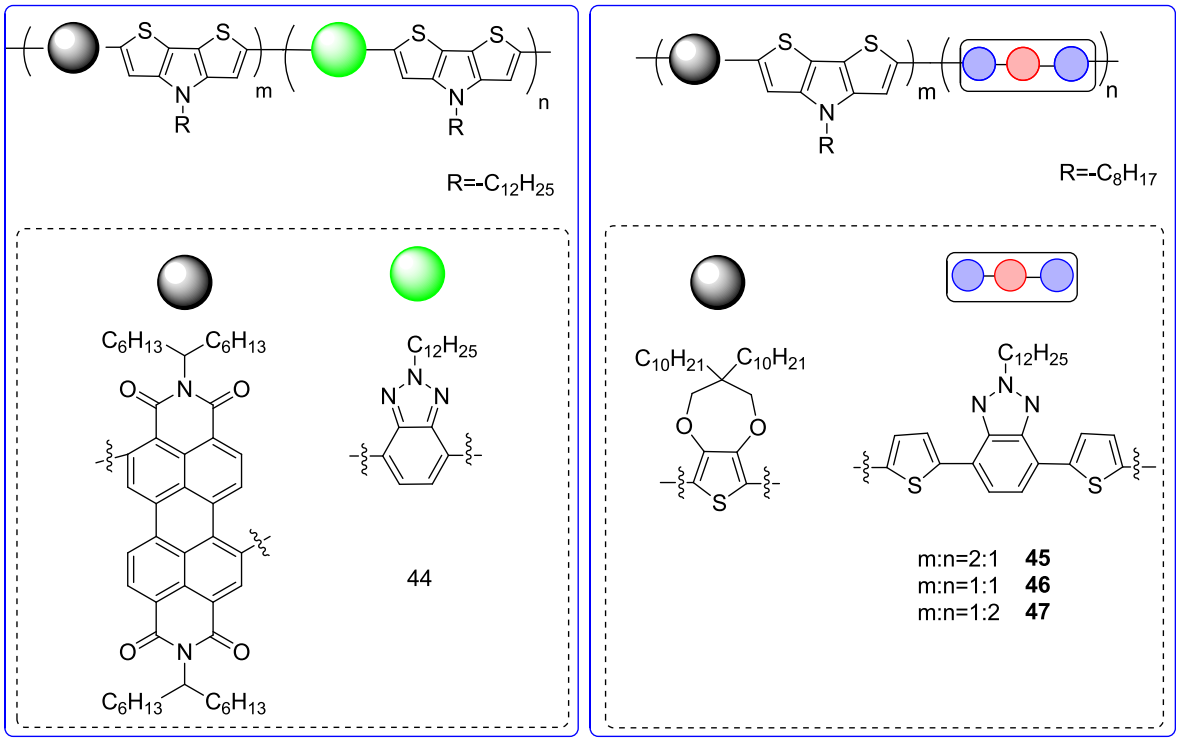
P *[ref]*	Synt. Meth.	λ_{abs} (nm) [b]	E_{gap} [c] (eV)	EC Properties [d,e] (L/a/b)
35 [57]	**E** **1:3**	450	–	*Showing EC property.*
36 [58]	**E** **1:1**	495	1.42*	*showing EC property*
37 [58]	**E** **1:4**	500	1.65*	*showing EC property*
38 [59]	**E**	470	1.77*	$t_{?\%}$[c] (s) = 1.54 (470 nm) / 1.47 (1216 nm), $t_{?\%}$[b] (s) = 1.95 (470 nm) / 1.92 (1216 nm), ΔT% =43.3(470 nm) / 63.5 (1216 nm), CE(cm^2/C) = 260.7 (1216 nm), yellow(0.0 V) (78/17/25.4), green(1.0 V) (80.4/-4.08/-2.46), ,blue(1.40 V) (76.7/-2.80/-3.80)
39 [60]	**E**	472	1.91*	ΔT% =50.7 (1212 nm), CE(cm^2/C) = 232.4 (1212 nm), brass (0.0 V) (81.89/15.77/18.07), ,gray green(0.8 V) (85.48/-0.94/1.99), gray(1.0 V) (84.46/-1.93/-4.97), grayish blue(1.40 V) (82.93/-2.40/-7.18)
40 [61]	**E**	484	–	$t_{90\%}$[c] (s) = 1.85 (484 nm) / 1.96 (1302 nm) $t_{90\%}$[b] (s) = 1.88 (484 nm) / 1.92 (1302 nm) ΔT% =39.6(484 nm) / 53.8 (1302 nm), (0.0 V) (82.49/14.04/14.73), (0.6 V) (87.79/-4.73/3.79), (1.20 V) (84.85/-3.03/-1.81)
41 [62]	**E**	466	–	$t_{?\%}$[c] (s) = 1.82 (1314 nm), $t_{?\%}$[b] (s) = 2.84 (1314 nm), ΔT% =45.3(1314 nm), reddish yellow(-0.2 V) (75.99/19.7/31.83), yellowish green(0.4 V) (79.74/6.47/26.24), green(0.6 V) (79.64/-4.95/6.07), blueish green(0.8 V) (76.97/-1.64/-6.83), blue(1.0 V) (76.97/-1.64/-6.83)
42 [62]	**E**	426	–	$t_{?\%}$[c] (s) = 2.28 (930 nm), $t_{?\%}$[b] (s) = 2.78 (930 nm) ΔT% =43.5(930 nm), yellow(0.2 V) (78.44/2.21/30.47), yellowish green(0.8 V) (78.99/-3.41/23.04), green(1.0 V) (73.98/-3.03/4.82), blueish green(1.20 V) (68.31/-0.68/-0.86)
43 [50]	**E**	436, 699	1.32	$t_{95\%}$[c] (s) = 1.30 (446 nm) / 4.43 (1100 nm), $t_{95\%}$[b] (s) = 8.1 (446 nm) / 1.72 (1100 nm), ΔT% =19(446 nm) / 41 (1100 nm), CE(cm^2/C) = 130 (446 nm) / 236 (1100 nm), gray-black (n), sky-blue (0.8 V), green blue (1. 0 V)
44 [63]	**C**	497	1.12	black (p-dop) (40.08/2.59/-1.037), brown (n) (50.68/2.418/37.78), blue (n-dop)(43.212/-48.87/-12.49), $t_{?\%}$ (s) = 1.6 (470 nm) / 0.9 (580 nm)/ 0.8 (1115 nm), ΔT% =9(470 nm) /10 (580 nm)/22 (1115 nm)
45 [64]	**C**	602	1.67	$t_{95\%}$[c] (s) = 0.85 (602 nm) / 1.78 (1580 nm), $t_{95\%}$[b] (s) = 0.84 (602 nm) / 0.44 (1580 nm), ΔT% =35.6(602 nm) / 58.1 (1580 nm), CE(cm^2/C) = 35.6(602 nm) / 58.1 (1580 nm), gray(-1.85 V), navy blue (0.0 V), light blue (1.40 V)
46 [64]	**C**	600	1.71	$t_{95\%}$[c] (s) = 1.04 (600 nm) / 0.88(1500 nm), $t_{95\%}$[b] (s) = 0.88 (600 nm) / 0.37 (1500 nm), ΔT% =30.8(600 nm) / 60 (1500 nm) CE(cm^2/C) = 94.3(600 nm) / 312.3 (1500 nm), (cm^2/C) = 94.3(600 nm) / 312.3 (1500 nm)gray(-1.80 V), navy blue (0.0 V), light blue (1.60 V)
47 [64]	**C**	598	1.73	$t_{95\%}$[c] (s) = 0.56 (598 nm) / 0.53(1570 nm), $t_{95\%}$[b] (s) = 0.63(598 nm) / 0.32 (1570 nm), ΔT% =40.7(598 nm) / 64.5 (1570 nm), CE(cm^2/C) = 265.2 (598 nm) / 400.6 (1570 nm), gray(-1.70 V), navy blue (0.0 V), light blue (1.60 V)

P: The code of polymer. **Ref**: reference **Synth. Meth.**: the type of synthesis method; **C:** chemical synthesis, **E**: electrochemical polymerization. λ_{abs}: absorption maximum for polymer.
[b] λ_{abs} in its neutral state.
[c] calculated from the onset of the π-π* transition of the neutral state polymer.
[d] the coloring switching time (at λ) at $t^c_{100\%}$: 100%, $t^c_{95\%}$: 95%, $t^c_{90\%}$: 95% of ultimate contrast, the bleaching switching time at $t^b_{100\%}$: 100%, $t^b_{95\%}$: 95%, $t^b_{90\%}$: 90% of ultimate contrast. If percentages or colored/bleached states (at a given λ) are not written, time is only given as t%.
[e] colors at a given applied potential or states; n: neutral state, i: intermediate state, ox: fully oxidized state.
* predicted data from reference given, red1: first reduced state, red2: second reduced state, ox1: first oxidized state, ox2: second oxidized state.

caused to decrease the intensity of band gap (1.12 eV). Furthermore, optical studies showed that ΔT% levels of the copolymer (**44**) were lower than that of other DTP-based copolymers as shown in Table 5**.** Three D-A typed soluble EC random copolymers (**45**, **46**, **47**) with different feed molar ratios were chemically synthesized by Zhao and coworkers [64] (Scheme 8). DTP structures were used as building blocks in the structure with 3,4-propylenedioxythiophene (ProDOT) as donors and this formation was cooperated with benzotriazole typed ternary structure. The experimental results showed that these copolymers had ambipolar characteristics (double doping and possessed double doping EC property colors (gray (during n-doping), navy blue (in the neutral state) and light blue (during p-doping)) upon oxidation with high ΔT% and CE and fast switching time values in NIR regions (Table 5).

2. Conclusions

This review mainly focuses on electrochromic properties of DTP-based conjugated organic polymers. The use of DTP monomer is advantageous for electrochromic applications as it allows tuning of optoelectronic and electrochemical properties via various structural modification possibilities compared to the discussed popular cores of electrochromic applications. This study gathers various modifications with large number of functional groups and side chains. These modifications and their effects are discussed covering many distinct polymers with a fine classification such as: DTP-based homopolymers, DTP-based alternating D-A type copolymers, DTP-based periodic D-A type copolymers and DTP-based random D-A type copolymers. The chemical structure modification of DTP provides the opportunity to adjust band gap for polymer and thus the polymer color can be tuned with respect to the conjugation and co-units added into DTP unit. Following the



Scheme 8. Chemical structures of DTP-based EC random copolymers-2 in literature.

pioneering work of Zotti et al. and later Rasmussen et al., EC DTP-based polymers began to attract attention. Research and developments in the EC field of polymers and copolymers obtained with N-functional DTP derivatives show that such polymers will become much more popular within the conjugated organic polymer family. The structurally flexible DTP platform serves as a starting monomer for the synthesis of a series of N-functional DTP-based homopolymers and copolymers with desired electrochemical and optical properties. To our regret, solubility of electropolymerized DTP structures is a problem because it restricts processability and the range of applications. When synthesizing DTP EC homopolymers, unfortunately, not much success can be achieved in overcoming solubility problems by adding functional units. In recent studies, attempts are being made to eliminate this problem by coupling the DTP structure with various conjugated monomeric structures in order to obtain polymers with the desired properties. With such an idea, it will be possible to synthesize new DTP-based EC polymers for EC applications in future.

CRediT authorship contribution statement

Buket Bezgin Carbas: Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization. **Hüseyin Bekir Yildiz:** Writing – review & editing, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

The authors dedicated this work to the 100th anniversary of the Republic of Turkey. On the 100th anniversary of our Republic, we commemorate our Leader Gazi Mustafa Kemal Atatürk, his companions who contributed to the establishment of the Republic, and all our heroes who worked to reach the 100th anniversary of the Republic, with mercy and gratitude.

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Buket Bezgin Carbas joined Professor Dr. Ahmet M. Önal's research group at Middle East technical University (METU), where she gained a Ph.D. for her work on electrochromic polymers and their applications in 2013. As a research scientist, between 2011-2013, she worked at the University of California Los Angeles (UCLA), Department of Materials Science and Engineering, in the State of California, USA. She worked on rechargeable batteries and polymer membrane gel electrolytes under the consultancy of Professor Dr Qibing Pei. She is presently working as a Professor in the Department of Energy Systems Engineering, Karamanoglu Mehmetbey University. Her main field of study is the synthesis of conjugated polymer derivatives and the application areas of the obtained conductive polymers. She has done many qualified studies on electrochromic and photoelectrochromic devices, solar cells, light-emitting diodes, biosensors, sensor polymers and capacitors. She has brought new potential members to the electrochromic materials family by synthesizing new polymers that can be used in the construction of electrochromic devices, especially in the field of materials science and energy.

Huseyin Bekir Yildiz completed his B.S and M.S in Department of Chemistry from Middle East Technical University, Ankara, Turkey and completed his Ph. D in Department of Chemistry, Middle East Technical University, Ankara, Turkey. Presently, he is working as a Professor in Department of Mechanical Engineering, Bartin University, Turkey. He was an active member who participated in many National and International conferences. His research areas are synthesis of DTP based electrochromics polymers, synthesis, characterization and applications of nanoparticles, solar cell and solar energy, biological photovoltaics, construction of novel biosensors, photo electrochemistry and electrochemistry.